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Title: Preliminary experimental tests and numerical modelling of an innovative base isolation system.

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Abstract

Seismic hazard is undoubtedly one of the most important causes of damage on civil structures and, very often, it is the cause for the loss of lives and goods as recently showed during the 7.8 M_W earthquake that struck southern and central Turkey and northern and western Syria, in February 2023. In the last decades, several generations of researchers devoted their efforts to study the most suitable seismic protection strategies, and, amongst others, seismic isolation has been proved to be one of the most effective and reliable [1].

Seismic isolation is a method of protecting a building from major earthquakes by installing some isolation and energy absorbing devices between the building superstructure and its foundation. Nowadays, many different isolation devices are present in the market with different mechanical properties and costs.

The present work reports the results of a preliminary experimental campaign carried out taking advantage of the 6-DOF shaking table system of the L.E.D.A. (Laboratory of Earthquake engineering and Dynamic Analysis) Research Institute at the “Kore” University of Enna, Italy [2]. The tests activities concerned an innovative seismic isolation system, named and patented as Double Raft, consisting of two foundation slabs in reinforced concrete. The lower slab made integral with the shaking table, can contain the upper one inside, on which a 1:2 scale steel frame with two elevations has been connected.

Two layers of Teflon sheets each 5 mm thick are interposed between the two foundations, to decrease the grazing friction between the two slabs and thus create a seismic isolation system at the base. To limit lateral movements, a system of steel springs was interposed between the walls of the lower slab and the upper slab. Upon assembly, the springs were preloaded to offer stiffness in both directions of the horizontal components of the relative displacement between the two slabs.

Random noise and 3D seismic experimental tests have been applied to the system under investigation and the experimental data are elaborated to develop an analytical model of the system, to verify the functioning of the proposed isolation system and to assess the seismic protection level offered.

References

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